GLOBAL STEWARDSHIP & RECONCILIATION TRUST (GSRT)

Citations & Evidence Addendum

Published sources behind the framework's key figures

Neil J. Schwab · Framework Architect · October 2026 · Supersedes the February 2026 edition

How to read this addendum. Each figure below is matched to a published source a reader can find. Where the literature gives a range, this addendum states which end the framework uses. Figures with no published source yet are labelled **Modelled assumption**.

No GSRT, MASH or LRP cohort has run. Every modelled figure will be replaced by measured pilot data. The simulations (Marcus, Laura, the 200-person cohort) are composite, illustrative cases, not field results.

Quick reference

Claim	Framework value	What the published sources show	Status
Annual public cost of a high-needs person	$120,000 (Observed) $80,000 (Conservative)	$29,600–$59,000 a year across five Canadian cities; about $56,000–$59,000 in the largest cities [1] $72,444 transient and $134,642 chronic, Calgary [3]	$120,000 fits the chronic, highest-need group only. Add an **Evidence Floor of $56,000**.
Savings from intervening early	MASH break-even analysis	$9.60 saved per $10 of Housing First for high-needs participants; $21.72 per $10 for the top 10% of service users [2]	Supports the logic that high users are where savings concentrate.
Land value from greening	30%	~6% nearby, peer-reviewed [4] 17% adjacent lots [5] Up to 30%, earlier summary [6]	30% is the **upper** estimate. Model 6%–17%.
Flood damage reduction from wetlands	60%	29% rural and 38% urban flood-cost reduction from avoiding wetland loss, Ontario [7] Up to 60% fewer flood-threat events under maximum rejuvenation, India [8]	Use the Canadian 29% as the floor.
Prevention vs. disaster relief	5× cheaper	$6 saved per $1 of hazard mitigation grants [9]	Supported; 5× sits below the published figure.
Worker sick days	12.5 days a year	Full-time health care employees: 18.0 days a year in 2022 [10]	Conservative; below the published figure.
Worker turnover	20% a year	Health care voluntary departures ~1.4% a month, about 17% a year [10]	Close; slightly above the published rate.

Claim	Framework value	What the published sources show	Status
LRP programme return	$8.28 per $1 (Observed) $5.52 per $1 (Conservative)	Canadian workplace mental health programmes: median $1.62 per $1 in year one; $2.18 after three years [11]	**Modelled assumption**, well above the published benchmark. Lead with the benchmark.
VVV multiplier (3.16875)	Designed constant	No published source; it is a design choice	Not evidence. Pilot calibration pending.

1. The public cost of a high-needs person

The best Canadian evidence puts the annual public cost of a homeless person with mental illness at roughly $56,000 to $59,000 in large cities ($29,600 in Moncton). The framework's $120,000 is defensible only for the chronic, highest-need group.

- **Five-city study (strongest source).** Latimer et al. followed At Home/Chez Soi participants in Vancouver, Winnipeg, Toronto, Montréal and Moncton. Annual service costs (2016 dollars, before medications) were $53,144, $45,565, $58,972, $56,406 and $29,610. Including medications, about $59,000 in the three largest cities. Costs covered health care, police, courts, incarceration, shelter and supportive housing, and income assistance. Individual costs ranged up to $341,535 [1].
- **Calgary estimates.** $72,444 a year for a transiently homeless person and $134,642 for a chronically homeless person (2008 figures, cited in Gaetz 2012) [3].
- **British Columbia.** $55,000 or more a year per homeless person (Patterson et al. 2008, cited in Gaetz 2012) [3].
- **Monthly unit costs, Toronto.** Shelter bed $1,932; provincial jail $4,333; hospital bed $10,900; rent supplement $701 [3].

Recommended three-track reporting. Observed Floor $120,000 (chronic, highest-need) · Conservative Floor $80,000 · **Evidence Floor $56,000** (large-city figure from the five-city study). At the Evidence Floor, 600 MASH participants represent $33.6M a year in public cost; MASH's full Year 1 cost of $11.25M (600 three-month stays at $7,500 through the 20/40/20/10/10 waterfall) breaks even if it avoids **33.5%** of that cost; at the $80,000 Conservative Floor, **23.4%**.

Why intervention pays where costs concentrate

The At Home/Chez Soi trial found that every $10 spent on Housing First for high-needs participants saved $9.60 in other services, and $21.72 for the 10% of participants with the highest service use at entry [2]. MASH targets the same high-use group, which is why screening for eligibility matters to the savings case.

2. Land value from greening

Greening vacant land raises nearby property values, but the peer-reviewed estimate is about 6%, not 30%.

- **Peer-reviewed.** Heckert and Mennis (2012) studied Philadelphia LandCare lots with a spatial difference-in-differences design. Nearby residential values rose about 6% more than near untreated lots, with effects strongest in moderately distressed areas [4]. (The 6% converts the paper's log-price coefficient of 0.056.)
- **Wharton / Federal Reserve Bank of Boston.** Wachter, Gillen and Brown (2008) report a 17% gain for homes adjacent to a stabilized, greened lot [5].

- **Earlier summary.** A 2005 University of Pennsylvania news summary of Wachter's work reported a 30% rise for neighbours of greened lots [6].

Correction. The February edition called 30% "the floor of the published range" and "a minimum." It is the highest of the three estimates. Model land value at 6% to 17%; cite 30% only as an upper bound.

3. Flood damage and prevention

Canadian evidence supports a 29% to 38% cut in flood costs from keeping wetlands, and U.S. evidence supports about $6 saved for every $1 spent on mitigation.

- **Ontario (most relevant).** The Intact Centre on Climate Adaptation (University of Waterloo, 2017) found that avoiding wetland loss would lower the cost of a major flood by 29% in rural areas and 38% in urban areas [7].
- **India, peer-reviewed.** Gupta et al. (2024, Journal of Hydrology) modelled a large river catchment. Under maximum wetland rejuvenation, flood-threat events at prominent cities fell by up to 60%; peak streamflow at the catchment outlet fell by 1–2% with existing wetlands and up to 8% with maximum rejuvenation [8].
- **Prevention return.** The National Institute of Building Sciences (2019) found that hazard mitigation grants save about $6 for every $1 spent [9]. The framework's "5× cheaper" sits below this figure.

Correction. The February edition used 60% (the top of the range, and a maximum-rejuvenation scenario) while describing it as conservative. Use 29%, the Canadian, cost-based figure, as the floor.

4. Worker burnout inputs (LRP)

Two of the LRP inputs are below published Canadian figures, one is close, and two still need a source.

Input	Framework value	Published figure	Status
Sick days per worker	12.5 a year	18.0 a year, health care employees, 2022 [10]	Conservative
Annual turnover	20%	~17% a year (1.4% a month) [10]	Close
Replacement cost	$93,000 (1.5 × $62,000 salary)	No source located	Modelled assumption
Mental injury claim	$15,000	No source located	Modelled assumption
Programme return	$8.28 per $1	Median $1.62 per $1; $2.18 after 3 years [11]	Modelled; above benchmark

Recommendation. Present Deloitte's published $1.62 to $2.18 benchmark first, then the LRP model as a projection the pilot will test. A funder trusts a model more when it is shown beside the real-world benchmark.

5. The VVV multiplier

The VVV multiplier (3.16875) is a designed integrity constant, not a statistically derived value and not evidence. It will be calibrated against real data in the pilot. The entropy calculation built on it is an

internal illustration. The Kenya landslide case is an illustrative simulation, not a field validation. Taleb's Antifragile (2012) is a published book, not a peer-reviewed study.

Every change from the February 2026 edition, and the reason for it, is recorded in Appendix B. Appendix A traces each derived figure back to its equation and inputs.

References

[1] Latimer E, et al. Costs of services for homeless people with mental illness in 5 Canadian cities: a large prospective follow-up study. CMAJ Open. 2017;5(3):E576–E585. doi:10.9778/cmajo.20170018. https://www.cmajopen.ca/content/5/3/E576

[2] Mental Health Commission of Canada. Cross-Site At Home/Chez Soi Project: Final Report, Executive Summary. https://mentalhealthcommission.ca/wp-content/uploads/2021/09/At2520Home2520Final2520Report2520-2520Exec2520Summary2520-2520ENG_0.pdf

[3] Gaetz S. The Real Cost of Homelessness: Can We Save Money by Doing the Right Thing? Toronto: Canadian Homelessness Research Network Press; 2012. (Cites Calgary Homeless Foundation 2008; Patterson et al. 2008; Shapcott 2007.) https://yorkspace.library.yorku.ca/handle/10315/29369

[4] Heckert M, Mennis J. The economic impact of greening urban vacant land: a spatial difference-in-differences analysis. Environment and Planning A. 2012;44:3010–3027. doi:10.1068/a4595. https://doi.org/10.1068/a4595

[5] Wachter SM, Gillen KC, Brown CR. Green Investment Strategies. Communities & Banking, Federal Reserve Bank of Boston. Spring 2008. https://www.bostonfed.org/-/media/Documents/cb/PDF/Wachter_greening.pdf

[6] University of Pennsylvania (Penn Today). Greening boosts home prices—here's the proof. February 24, 2005. https://penntoday.upenn.edu/2005-02-24/research/greening-boosts-home-prices%E2%80%94here%E2%80%99s-proof

[7] Intact Centre on Climate Adaptation, University of Waterloo. When the Big Storms Hit: The Role of Wetlands to Limit Urban and Rural Flood Damage. 2017. https://uwaterloo.ca/water-institute/news/conserving-wetlands-could-save-canadians-millions-flood

[8] Gupta R, et al. Assessing the efficacy of hydro-ecological based wetland management approach for flood resilience of a large river catchment. Journal of Hydrology. 2024;641:131761. https://ui.adsabs.harvard.edu/abs/2024JHyd..64131761G/abstract

[9] National Institute of Building Sciences. Natural Hazard Mitigation Saves: 2019 Report. Washington, DC; 2019. https://www.preventionweb.net/publication/natural-hazard-mitigation-saves-2019-report

[10] Statistics Canada. Study: Quality of employment of health care workers during the COVID-19 pandemic. The Daily, August 10, 2023. https://www150.statcan.gc.ca/n1/daily-quotidien/230810/dq230810a-eng.htm

[11] Deloitte Canada. The ROI in workplace mental health programs: Good for people, good for business. November 2019. https://www.insurance-canada.ca/2019/11/11/deloitte-research-workplace-mental-health-return-on-investment/

Appendix A — Equation Trace

Each figure used in partner documents is recomputed from its equation. The trace checks the arithmetic; the input status shows where published evidence ends and design choices or modelled assumptions begin. An equation can confirm that a result follows from its inputs. It cannot confirm the inputs; only a source or pilot data can.

Input status key. Sourced = a published reference in this addendum · Design = a locked framework choice · Modelled = an assumption the pilot will test · Unsourced = no basis located.

Figure	Equation	Inputs and status	Result	Check
MASH break-even, Evidence Floor	Gross cost ÷ (participants × cost per person)	$11.25M (Design) · 600 (Design) · $56,000 (Sourced [1])	$11.25M ÷ $33.6M = 33.5% (43.1% with returns)	Arithmetic confirmed
MASH break-even, Conservative	Same	$80,000 (Modelled)	$11.25M ÷ $48.0M = 23.4% (30.2% with returns)	Arithmetic confirmed
MASH break-even, Observed	Same	$120,000 (Sourced for chronic, highest-need only [3])	$11.25M ÷ $72.0M = 15.6% (20.1% with returns)	Arithmetic confirmed
MASH Year 1 cost	600 × $7,500 ÷ 0.40 (stays are the 40% stream)	$7,500 per 3-month stay (Design): 90 × $80 + 3 × $60 = $7,380	$4.5M stays → $11.25M gross; about $14.5M with return stays	Resolved: replaces the unitemized $6.5M
MASH pilot ask (old documents)	Sum of budget lines	Eight budget lines (Design)	$3,750,000	**Old $3.5M ask was $250K short; superseded by $11.25M**
Earthworks wage	Hours × days × weeks × rate	6 h · 5 d · 50 wk · $22 (Design, locked)	1,500 h × $22 = $33,000	Arithmetic confirmed
Earthworks gross revenue required	R = Payroll ÷ 0.40	200 × $33,000 (Design) · 0.40 (Firmware)	$6.6M ÷ 0.40 = $16.5M	Arithmetic confirmed
Justice Floor	20 + 40 + 20 + 10 + 10	Firmware (Design)	100%	Sums correctly
Earthworks vs status quo	$16.5M vs 200 × cost per person	$120K / $80K / $56K tracks	$24.0M / $16.0M / $11.2M	**At the Evidence Floor Earthworks costs $5.3M more unless contract revenue covers it**
LRP current cost	(T + S + P + H + C) × n	$10,000 + $800 + $1,500 + $1,000 + $2,000 (Modelled; S, T partly Sourced [10]) · n = 20	$15,300 × 20 = $306,000	Arithmetic confirmed
LRP return	ΔCost ÷ programme cost	25% reduction (Modelled) · $9,238 (Design)	$76,500 ÷ $9,238 = 8.28; payback 44.1 days	Arithmetic confirmed; benchmark $1.62–$2.18 [11]
LRP return, Conservative	Same, inputs × 2/3	As above	$51,000 ÷ $9,238 = 5.52	Arithmetic confirmed

Figure	Equation	Inputs and status	Result	Check
LRP burnout exposure	(W × T_r × $93,000) + (S_d × W_d × W) + (A_r × A_c) + M_i	$93,000 and $15,000 (Unsourced) · A_r = 4 · others Modelled	$372,000 + $59,500 + $10,000 + $15,000 = $456,500	**Reconciled: the written formula lacked × W and A_r = 4. Exposure ceiling only; the 49.4× ratio assumes 100% avoided and is not cited**
W score (Marcus)	(R + B + N + V + M + Gv + Fe + Fm) × Cs	Sub-scores from a composite case (Modelled) · Cs 0.8 / 1.0 / 1.2 (Design)	14 × 0.8 = 11.2 → 48 × 1.0 = 48.0 → 67 × 1.2 = 80.4	Confirmed; **Years 1–2 use Cs 1.1 and 1.15, outside the locked set**
Seed repayment	Return = ROI River × 10	×10 multiplier (Unsourced)	$37M cumulative gross	**Cannot verify; do not cite until sourced**
Sun Vow on-territory capital	Gross × retention share	$38.5M gross (Unsourced) · 80% / 95% (Design)	$30.8M vs $36.6M; + $5.8M	Arithmetic confirmed; **gross needs a headcount basis**

Appendix B — Record of Changes

Changes from the February 2026 edition to this October 2026 edition, with the reason for each.

Item	February 2026	October 2026	Why
Public cost per person	$120,000, composite of agency figures with no report or table named	Three tracks: $120,000 / $80,000 / $56,000 Evidence Floor	The agency citations could not be traced; the five-city peer-reviewed study gives $56,000–$59,000 in large cities [1]
Land value from greening	30%, called “a minimum” and “the floor”	6%–17% modelled; 30% as upper bound	The source said “as much as 30%”; peer-reviewed estimate is about 6% [4], Wharton 17% [5]
Flood reduction	60%, called conservative	29% (Ontario) as floor	60% is the maximum-rejuvenation scenario in one Indian catchment [8]; a Canadian cost study exists [7]
Prevention return (5×)	Backed by “UNDRR $6–$10 per $1” (not located)	Backed by NIBS 2019, $6 per $1	A primary source was found [9]
VVV multiplier	“Verified”; entropy result “verified by Gemini AI system review”	Designed constant; internal illustration	An AI review is not a source; the canon labels VVV a designed constant
Kenya case	“Field-validated”; $2.93B cascade value	Illustrative simulation; figure removed	The canon register labels Kenya a simulation, never field validation
Per-person swing	$185,000 swing; “exceeds $2 billion annually”	Removed	Adds avoided cost and wages together; $65,000 does not match the locked $33,000 wage
LRP return	$8.28, “actual savings likely higher,” “H&S Board data with half-value inputs”	Modelled projection shown beside the $1.62–$2.18 benchmark	No source located for the H&S Board claim; Deloitte gives the real-world benchmark [11]
Worker sick days and turnover	Unsourced	Sourced to Statistics Canada	18.0 days and ~17% a year for health care workers [10]
Antifragility (Taleb)	“Published & peer-reviewed”	Published book	Antifragile (2012) is a book, not a peer-reviewed study
Secondary sources	Coldwell Banker Bahamas page; India Water Portal summary	Replaced by primary studies	A real-estate blog and a news summary are not evidence for a funder
Closing statement	“The real numbers will almost certainly be higher”	Removed	The pilot decides that
MASH stay and cost	$7,500 labelled as a 4-month stay	$7,500 per 3-month stay, with a one-month shadow overlap inside it	$80/day + $60/month = $7,380 for 90 days; locked by the Framework Architect, 18 Oct 2026
MASH Year 1 ask	$3.5M (budget lines sum to $3.75M)	$11.25M gross (about $14.5M with returns)	600 stays through the full 20/40/20/10/10 waterfall
MASH break-even	19.3% on an unitemized $6.5M	33.5% Evidence Floor · 23.4% Conservative · 15.6% Observed	Recomputed on the full waterfall cost

Item	February 2026	October 2026	Why
LRP 10-year value	$918,000 (Mathematics Manual v2, §5.1)	$765,000 with no ripple factor	$76,500 × 10 = $765,000; $918,000 assumes an unstated ripple factor of 2
LRP return wording	$8.28 “returned”	$8.28 saved per $1 (gross); $7.28 net	Net return subtracts the programme cost
Burnout exposure (E_ab)	Formula omitted × workforce on sick days	(W × T_r × $93,000) + (S_d × W_d × W) + (A_r × A_c) + M_i, with A_r = 4	Reproduces the stated $456,500; exposure ceiling only, never used in ROI
Edition date	February 2026	October 2026	Supersedes all earlier copies

FORGIVE YESTERDAY · HEAL TODAY · FLOURISH TOMORROW · TOGETHER